

COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

*Machine-computed from 16 years of daily price data through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. **These numbers are computed, not judged, so they carry no draft/committed status of their own — they enter the master blend directly. This cycle's active qualitative score set is 2026-09-16_draft, a light daily touch — CLARITY Act re-rated DOWN on the failed cloture vote, Iran War Risk re-rated down modestly on the nuclear track, Tokenization and Fed & Macro held flat — COMMITTED and AUTO-STAMPED. See those pillar reports for the specific re-rates. This quant pillar's own output is unaffected by qualitative status and enters the master blend the same way regardless.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through September 17, 2026 (BTC \$76,440.74) · Pipeline re-run via a full LOCAL run.py quant pass & verified September 17, 2026 (qualitative set: light daily touch, 2026-09-17_draft, committed — auto-stamped) · Not Financial Advice

The Scores — September 16 Model Output

Computed September 17, 2026 (previous computed run September 16, 2026; this cycle ran a full LOCAL run.py quant pass — a real Bitstamp/Coin Metrics fetch, no data-relay workaround needed — then indicators/levels/scores/cycle/mapping/charts recomputed locally) — same standard as every qualitative pillar. This cycle no new weekly, monthly or quarterly bar closed, so the completed-bar composites are unchanged; the current bars are PARTIAL and both readings are quoted together throughout.

WINDOW	MODEL P	QUALITY	IMPACT	SIGNAL	CALL
1 month	58	0.75	8	+0.96	Mild Bull — the weekly close above the 200-week MA remains confirmed; the still-forming week reads +30.4 as-is, +31.3 on completed bars; 1mo analog 2-of-3 positive, median +5.0%
3 months	54	0.70	8	+0.45	Neutral — still the softest window; the 3mo analog reads 2-of-3 positive, median +13.7%
6 months	59	0.55	8	+0.79	Mild Bull — unanimous 3-of-3 positive, median +45.7%
1 year	69	0.60	8	+1.82	Bull — the cycle turn; every prior cycle at this exact month was higher a year later (3-of-3), median +97.8%
3 years	69	0.50	8	+1.52	Bull — value + cycle (36-month analog 3-of-3 positive, median +1395.4%)

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** Signal = ((P – 50) ÷ 50) × Quality × Impact · **Honesty cap:** P = 50 + 25 × (window score ÷ 100) — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

- P** — the odds the tape is *good for Bitcoin* by then. Near-term still confirmed on a completed-bar basis, and the still-forming week reads \$76,440.74 (weekly composite as-is +30.4/100, Slight bullish, agreement 5-of-7; completed-bar weekly composite holds at +31.3); long-term still high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, median +97.8%, and price sits ~46% below its power-law fair value).
- Quality** — how trustworthy the evidence is. Highest near-term (843 weekly bars of history feeding the indicator stack), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).
- Impact** — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

- The crowd:** the Fed's Sept 16 rate hike and hawkish dot plot dominate today's headlines — "tightening for longer, the tape should reflect it." (Price sits at \$76,440.74, up modestly from \$76,138.60 on Sept 16 — still +30.2% off the June 30 low; the tape's muted reaction is itself information: the hike was largely priced, and the incremental hawkish signal sits in the Fed & Macro pillar's qualitative re-rate, not in this pillar's tape-only read.)
- The model:** the short-term tape remains genuinely confirmed on a completed-bar basis and ticked up slightly with price — the weekly composite (as-is) reads Slight bullish (+30.4, agreement 5-of-7), the monthly composite reads Neutral (–1.4, agreement 2-of-7), and the quarterly composite reads Slight bullish (+24.0, agreement 4-of-7). No

new weekly, monthly or quarterly bar closed this cycle, so completed-bar composites are unchanged: weekly +31.3, monthly +1.8, quarterly +7.7. Repair Signal #1 (a weekly close above the 200-week MA) stays confirmed. The 1-month window eased to Mild Bull (P 58, from 60) as the cycle-month clock advanced to 28.94 and its analog slipped to 2-of-3 positive (median +5.0%); 3 months holds Neutral (P 54) with its analog at 2-of-3 positive (median +13.7%). The 6-month window holds Mild Bull (P 59) with the analog still unanimous (3-of-3 positive, median +45.7%). The long read held at 1yr and 3yr (both 69): every prior cycle at this exact month was higher 12 months later (median +97.8%), price is ~46% below fair value (\$140,453), and the froth that precedes real crashes remains absent.

The edge: the crowd reads one clock (the news) — and today's news is a hawkish Fed, already mostly priced by the market. The model still reads three separate clocks that don't take a position on FOMC dot plots — tape (confirmed at the weekly level), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 28.94, past the average-bottom date), and valuation (~46% below the power-law fair line, with price sitting above the -1σ "cheap" band at \$70,143). All three clocks point the same direction on the medium-to-long horizon — the near-term clock is the one doing the day-to-day flexing, exactly as its lower Quality input says it should.

The Facts — The Tape

*Computed September 17, 2026 from daily data (BTC \$76,440.74, daily close as of September 17). Current weekly/monthly/quarterly bars are still forming — scores use them as-is. Quantified: on completed bars only, the weekly composite is **+31.3** (current week closes 2026-09-20); monthly +1.8 and quarterly +7.7 (both bars closing 2026-09-30) are unchanged. Both readings are quoted deliberately — the as-is figure includes a bar that has not finished.*

- **Weekly composite +30.4/100 (Slight bullish), agreement 5-of-7 indicators** — ticked up with price; on completed bars only it reads +31.3, and both must be quoted together because the current bar is still forming. Price is above the 20-week (\$69,960, +8.9%) and well above the **200-week average (\$65,471, +16.4%)**; price sits below the 50-week (\$78,691, -3.2%). RSI 53.8; MACD line negative (-426) with histogram positive (+2,272) — momentum repair continuing; %B 0.72.
- ★ **Repair signal #1 (a weekly close above the 200-week MA) remains CONFIRMED.** Price (\$76,440.74) sits +16.4% above the 200-week average (\$65,471). No new weekly bar closed this cycle (the current week closes Sept 20), so the completed-bar weekly composite holds at **+31.3**, unchanged.
- **Monthly composite -1.0/100 (Neutral), agreement 2-of-7** — ticked up with price. Below the 20-month average (\$86,910, -12.3%), above the 50-month (\$62,109, +22.7%); RSI 49.9; monthly MACD line +132 with histogram -3,877 — still negative; %B 0.35. Completed-bar monthly composite (August's closed bar) holds at +1.8, unchanged.
- **Quarterly composite +24.0/100 (Slight bullish), agreement 4-of-7** — essentially flat with the cycle-month clock. Comfortably above the 20-quarter average (\$58,032, +31.3%) and the 50-quarter (\$28,689, +165.6%); RSI 56.7, MACD line +16,041 with histogram -582. Completed-bar quarterly composite is +7.7, unchanged (current bar dated 2026-09-30, still forming).
- **Levels that matter:** price (\$76,440.74) sits between the nearby \$74,434 support (-2.3%, 1 touch) and overhead resistance at \$81,685 (+7.2%, 2x touch) — the nearest test on the board — then \$89,164 (+17.0%, 1x) and \$98,090 (+28.7%, 2x). Further below, \$59,930 (-21.4%, 1x) and \$57,735 (-24.2%, 1x). The falling weekly support trendline sits at \$41,829; the falling resistance trendline sits at \$74,733, just below current price.
- ★ **June 30 remains the cycle low: \$58,526 close** — the flush tagged the Fib golden pocket (\$54,257–\$57,802 = 61.8–65% retrace of the \$15,479 → \$126,272 run) and has since built a **+30.2% bounce to \$76,440.74. \$58,526 is still the line in the sand** — a decisive break back below it would undo this entire recovery leg.



Weekly tape, scored +30.4/100 as-is (Slight bullish, +31.3 on completed bars only) with agreement 5 of 7 indicators: the weekly close above the 200-week MA remains confirmed, MACD histogram remains positive. Machine-annotated from the same data that feeds the scores.

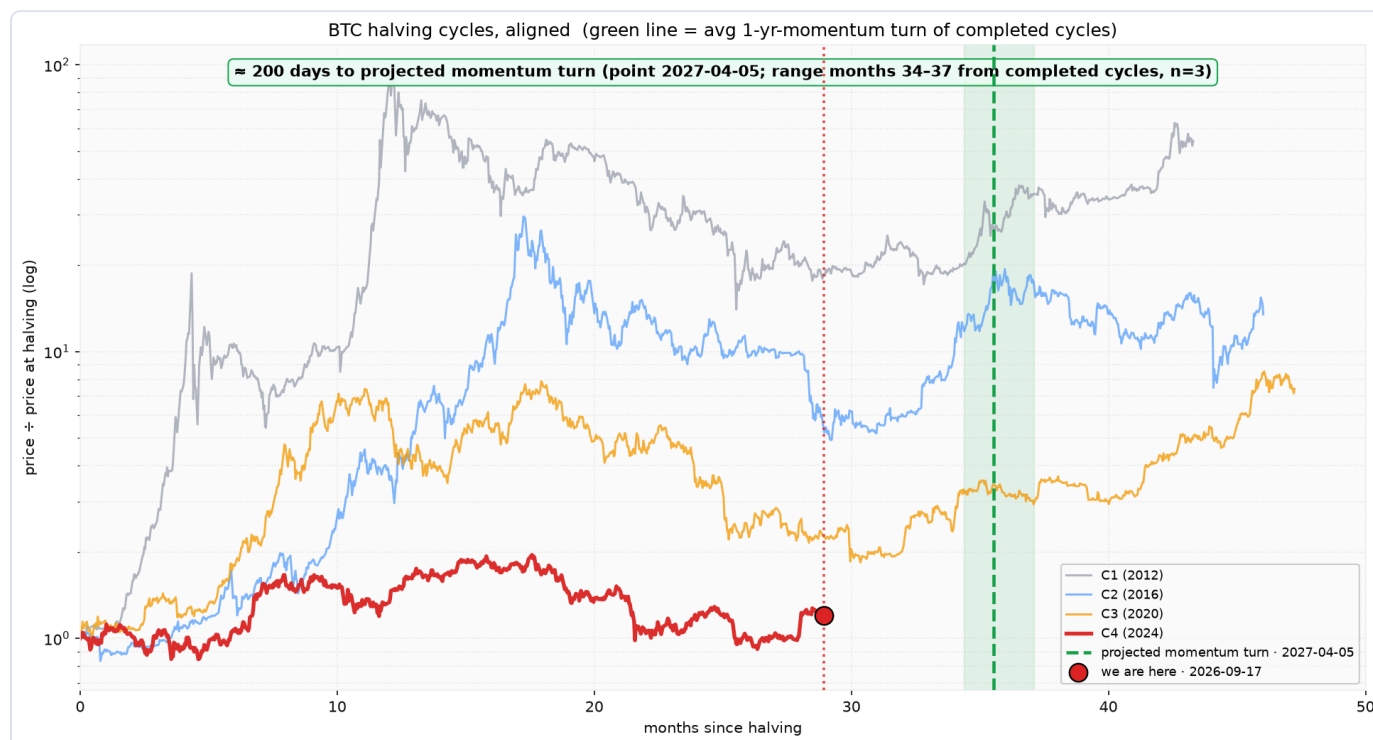
The Facts — The Cycle Clock & The Value Floor

The two slower clocks. Same data, longer lenses.

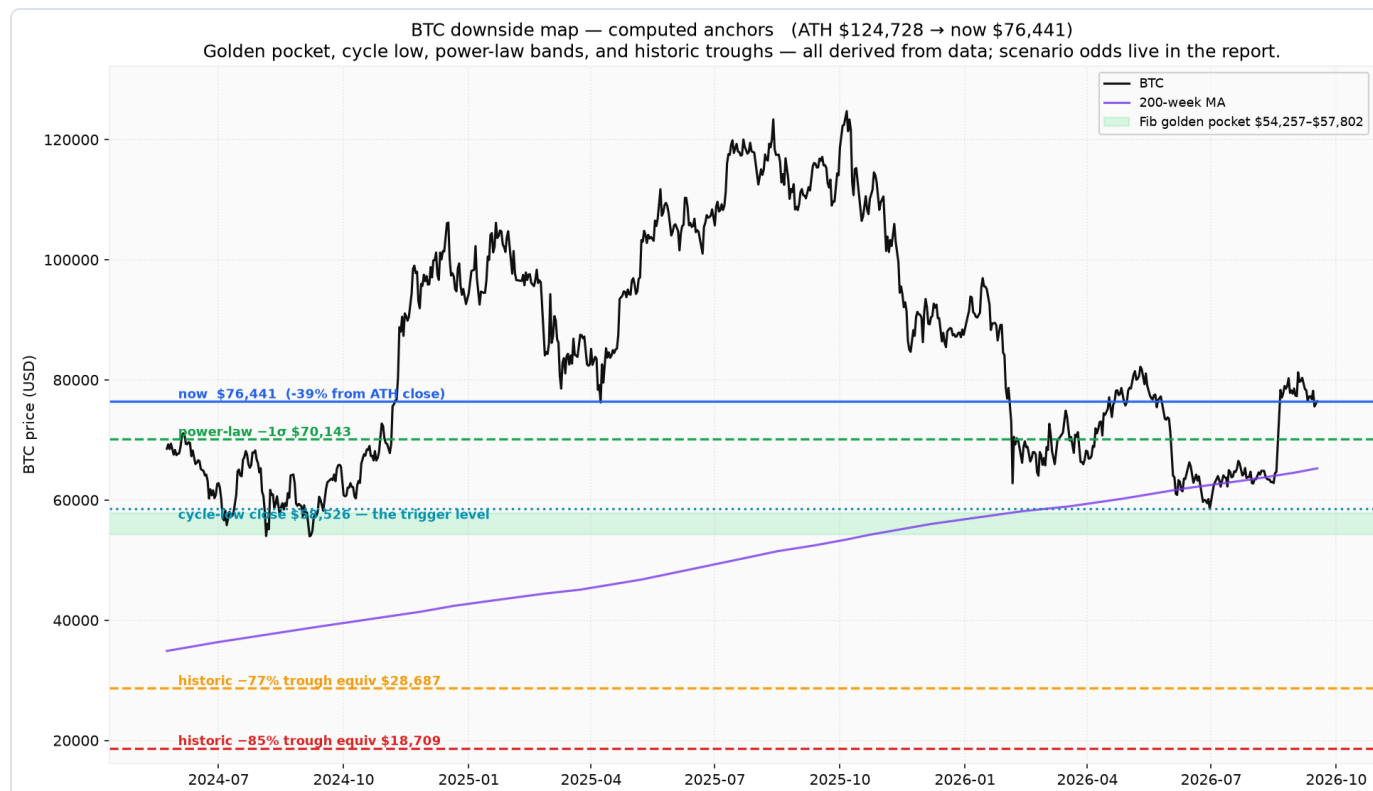
- The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving (avg 15.8) and bottomed at months 26–30 (avg 28.4). This cycle topped at month 18 (Oct 6, 2025 — \$124,728) and printed its low-so-far at month 26 (June 30, \$58,526, –53% from the top). **We are at month 28.94 — past the average-bottom date.**
- ★ The cycle analogs are the spine of the long-window scores.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive **2-of-3** (median +5.0%) · next 3 months positive **2-of-3** (median +13.7%) · next 6 months positive **3-of-3** (median +45.7%) · next 12 months positive **3-of-3** (median +97.8%) · next 36 months positive **3-of-3** (median +1395.4%). *Fragility disclosure: the medians and agreement counts move as the cycle-month clock advances — treat the analog levels as soft, the direction as the signal, and this is exactly the kind of alignment sensitivity this disclosure exists to flag. And know what the 36-month number is: from month ~29 that window reaches month ~65 — it contains the NEXT halving's rally.*
- 1-year momentum turned positive at months 34–37 in every completed cycle (avg 35.5) → still projects to ~April 2027 for this cycle.**
- The power-law floor:** a log-log fit over the full price series puts fair value at **\$140,453** — price is **–46%** versus fair. Fit-window honesty: starting the fit in 2011, 2012, 2013, or 2015 instead moves fair value materially (\$130,181 / \$135,106 / \$118,374 / \$145,224) — the level is soft, but the **"well below fair" verdict holds across every window tested**. Price (\$76,440.74) sits above the **–1σ band (\$70,143)** — out of the "cheap" zone (the –2σ band sits at \$35,030, far below). Every halving-era cycle trough was –77% to –85% from ATH (\$18.7K–\$28.7K

equivalent today) — the pre-halving 2011 bear went $\sim -93\%$ ($\sim \$8.7K$ equivalent), excluded from the range but in the data; this cycle's low-so-far was -53% from the ATH.

- **No froth in the system:** the altcoin-froth index reads $z = -0.87$ — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to -39% average 12-month drawdowns vs -23% for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~ 6 weeks (through May 23, unchanged)* — used as a tilt, not a score input.
- ★ **What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to $\sim$ April 2027 for this one. The red dot is today: month 28.94, past the historical bottoming window.



The downside map, updated after the June 30 flush: the projected \$54K–\$61K bottom zone was hit (\$58,526 close / \$57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to \$54–\$58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From \$76,440.74: the 20-week average (~\$70.0K) is well behind us — price sits **+8.9%** above it; the 50-week average (\$78,691) is still overhead as the immediate test, now **3.2%** away. A retest of the \$124,728 ATH is **+63.7%**; the power-law fair line is **+84.3%**. The median 12-month analog from this cycle-month is now **+97.8%**; the median 36-month analog is +1395.4% — **discount that one hard**, it leans on regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the near-term signal remains confirmed on a completed-bar basis and holds Mild Bull (1mo P 58) even as the Fed & Macro pillar re-rated down today on the FOMC's hawkish dot plot — the near quarter is still the softest window (3mo P 54, barely above a coin flip), but the 6-month-and-beyond case remains unanimous (3-of-3) and holds steady (6mo P 59).

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A -53% drawdown at month 26 vs -77/-85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is \$18.7K–\$28.7K, far below every level in this report.

- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.
- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.
- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
\$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket \$54.3–57.8K is the target; re-rate 1/3-month down
~\$65,470	Weekly close above the 200-week MA	CONFIRMED — price sits +16.1% above this level; completed-bar weekly composite holds at +31.3, unchanged this cycle (no new weekly bar yet, current week closes Sept 20)
~\$78,687	Weekly close above the 50-week MA	Price sits –3.4% below it — the next real test; a confirmed weekly close above it would extend trend repair further — re-rate 1/3-month up if it holds
~Aug 31, 2026	Historical average-bottom month (28.4) has now passed (we're at month 28.94)	Price sits well off the June 30 low (+29.9%); a higher low + improving monthly momentum after this date = basing confirmed — August's completed monthly bar (+1.8) is an early such sign
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
\$50,000	Weekly close below	–60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **\$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone (\$18.7K–\$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

QUALITATIVE SET: COMMITTED — AUTO-STAMPED (LIGHT DAILY TOUCH, 2026-09-17_draft) · THIS PILLAR: COMPUTED SEPTEMBER 17, 2026 (NO DRAFT/COMMITTED STATUS OF ITS OWN)

Computed P: 58 / 54 / 59 / 69 / 69 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8

This is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. No new weekly, monthly or quarterly bar closed this cycle (the completed-bar weekly composite holds at +31.3, unchanged), and the as-is weekly composite eased to +29.1 (5-of-7) with price. The 1-month window holds Mild Bull (P 60, down 1 from 61) as the cycle-month clock advanced to 28.91, with its analog unanimous (3-of-3 positive, median +6.0%). The 6-month window holds Mild Bull (P 59) with its analog still unanimous (3-of-3 positive, median +44.0%). The 3-month window ticked up slightly (P 54, from 53) — still the softest window in the set, its analog at 2-of-3 positive (median +12.2%). The long read ticked up at 1yr (68→69) and held at 3yr (69) — the direction, not the level, is the signal.

These numbers enter the master blend exactly as computed — this pillar has no override/re-stamp step of its own. Unlike the four qualitative pillars, this pillar carries no draft/committed status of its own: this cycle's active qualitative score set is 2026-09-16_draft, a light daily touch — CLARITY Act re-rated down on the failed cloture vote, Iran War Risk re-rated down modestly on the nuclear track, Tokenization and Fed & Macro held flat — COMMITTED and AUTO-STAMPED. This pillar's own computed output is unaffected by that process and enters the master blend the same way it does every cycle.

What Changed Since Last Check

Pipeline data refreshed September 17, 2026 via GitHub Actions (previous computed run September 16, 2026).

- **Today's refresh in one line:** a full local run.py quant pass (Bitstamp/Coin Metrics fetch + recompute, not a data-relay read) moved BTC from \$76,138.60 (Sept 16) to **\$76,440.74** (Sept 17), and the model P from 60/54/59/69/69 to **58/54/59/69/69**.
- **Data revision, disclosed.** Because this cycle fetched the exchange feed directly rather than reading the relayed snapshot, it also picked up revisions to the prior days' closes: **Sept 14 now reads \$78,185.01** (the relay had \$77,866.59) and **Sept 15 now reads \$75,577.29** (the relay had \$76,916.53). The Sept 16 and Sept 17 closes also differ slightly from the relay values the cloud run used (\$76,013.81 and \$76,197.81). Every figure in this report is the current local fetch, which is what the model actually scored; the superseded values are named here rather than quietly overwritten.
- **Price: \$76,916.53 (Sept 15) → \$76,138.60 (Sept 16) → \$76,440.74 (Sept 17).** A modest bounce, tracking the Sept 16 FOMC's muted market reaction, though this pillar scores the tape, not the news.
- ★ **Repair signal #1 remains CONFIRMED:** price sits +16.4% above the 200-week average (\$65,471). No new weekly bar closed this cycle, so nothing about that confirmation changed.
- **NO new weekly, monthly or quarterly bar closed this cycle.** Completed-bar composites are therefore unchanged: weekly **+31.3**, monthly **+1.8**, quarterly **+7.7**. The still-forming (as-is) composites ticked up with price: weekly **+30.4** (agreement 5/7), monthly **-1.0** (agreement 2/7), quarterly **+24.0** (agreement 4/7). Both readings must be quoted together because the current bars are PARTIAL.
- **Model P vs Sept 16: 60/54/59/69/69 → 58/54/59/69/69.** 1mo **-2**; 3mo/6mo/1yr/3yr flat. The move is entirely the cycle-analog clock, not the tape: as the cycle-month advanced to **28.94**, the 1-month analog now reads 2-of-3 positive (median +5.0%) and the 3-month analog holds 2-of-3 positive (median +13.7%). Honesty note,

unchanged: analogs rest on **n=3 completed cycles** and single windows flip when alignment shifts by days — the direction is the signal, the levels are soft.

- **Power-law fair value: \$140,453 (Sept 16) → \$140,453 (Sept 17)** (price now −46% vs. fair) — normal drift as new daily bars extend the fit. Fit-window sensitivity this cycle spans \$118,374 (2013 start) to \$145,224 (2015 start): quote it as a range, never a target.
- **Froth unchanged at $z = -0.87$** , still dated 2026-05-23 — Coin Metrics community data lags ~6 weeks, so this stays a robustness tilt, never a score input.
- **Data refreshed:** daily price data through September 17, 2026 via a full local `run.py` quant pass (direct Bitstamp/Coin Metrics fetch), then `indicators/levels/scores/cycle/mapping/charts` recomputed locally.
- **Context:** this cycle's active qualitative score set is **2026-09-17_draft**, a light daily touch, COMMITTED and auto-stamped, in which Fed & Macro re-rated DOWN modestly on the FOMC's hawkish dot plot, while CLARITY Act, Iran War Risk, and Tokenization held flat. This quant pillar is computed fresh every run regardless of qualitative status.

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via [fetch_btc_history.py](#) · Coin Metrics community reference price (2010–2011 close-only) via [build_master_history.py](#) — 5,838 days total, each row tagged by source.
- **Indicator stack:** [indicators.py](#) — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. [levels.py](#) — swing-cluster S/R + least-squares trendlines.
- **Scoring:** [score.py](#) — each indicator → conviction in [−100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** [cycle_analysis.py](#) — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · [downside.py](#) / [downside_map.py](#) — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · [test_altcoin_froth.py](#) — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** [pillar_mapping.py](#) — window scores from tape composites + cycle analogs + power-law value; $P = 50 + 25 \times (S/100)$; Quality rubric per window; writes `data/pillar_scores.json` for the master blend.
- **Run order:** `fetch_btc_history` → `build_master_history` → `fetch_altcoin_data` → `indicators` → `levels` → `score` → `charts` → `cycle_analysis` → `downside` → `downside_map` → `pillar_mapping`.
- **Honesty notes:** current weekly/monthly/quarterly bars are partial as of Sept 17 (completed-bar composites: weekly +31.3 / monthly +1.8, August's bar having closed / quarterly +7.7, current bar still forming — all three unchanged this cycle since no new bar closed) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · $n=3$ on all cycle claims, and analog windows can cross into the next cycle · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.